

STA 1020A QUIZ

a) A company uses three call centers (A, B, and C) to handle customer support, contributing 40%, 35%, and 25% of the total calls, respectively. The probabilities that a call is unresolved are:

- Call Center A: 4%,
- Call Center B: 3%,
- Call Center C: 5%.

i. What is the probability that a randomly selected call is unresolved?

[2 marks]

ii. If a randomly selected call is unresolved, what is the probability that it came from Call Center C?

[2 marks]

b) In a health survey, it was found that:

- 65% of people exercise regularly (E),
- 50% follow a healthy diet (D),
- 40% both exercise and follow a healthy diet.

If a person is selected at random, what is the probability that:

i. The person either exercises or follows a healthy diet (or both)? **[2 marks]**

ii. The person follows a healthy diet, but does not exercise? **[2 marks]**

iii. The person neither exercises nor follows a healthy diet? **[2 marks]**

c) A small bakery produces batches of cookies, where the number of defective cookies Y in a randomly selected batch follows the probability mass function:

$$f(y) = \frac{1}{15}(y+1), y = 0, 1, 2, 3, 4.$$

i. **Verify** that the given function is a valid probability mass function. **[2 marks]**

ii. Find the probability that a randomly selected batch contains at least 3 defective cookies. **[2 marks]**

iii. Suppose the bakery discards any batch with more than 2 defective cookies. What is the probability of discarding a randomly selected batch? **[2 marks]**

iv. Calculate the expected number of defective cookies, $E(Y)$. **[2 marks]**

v. Calculate the variance of the number of defective cookies, $\text{Var}(Y)$. **[2 marks]**